

BALL AND PLAYERS TRACKING IN PADEL MATCHES VIDEOS

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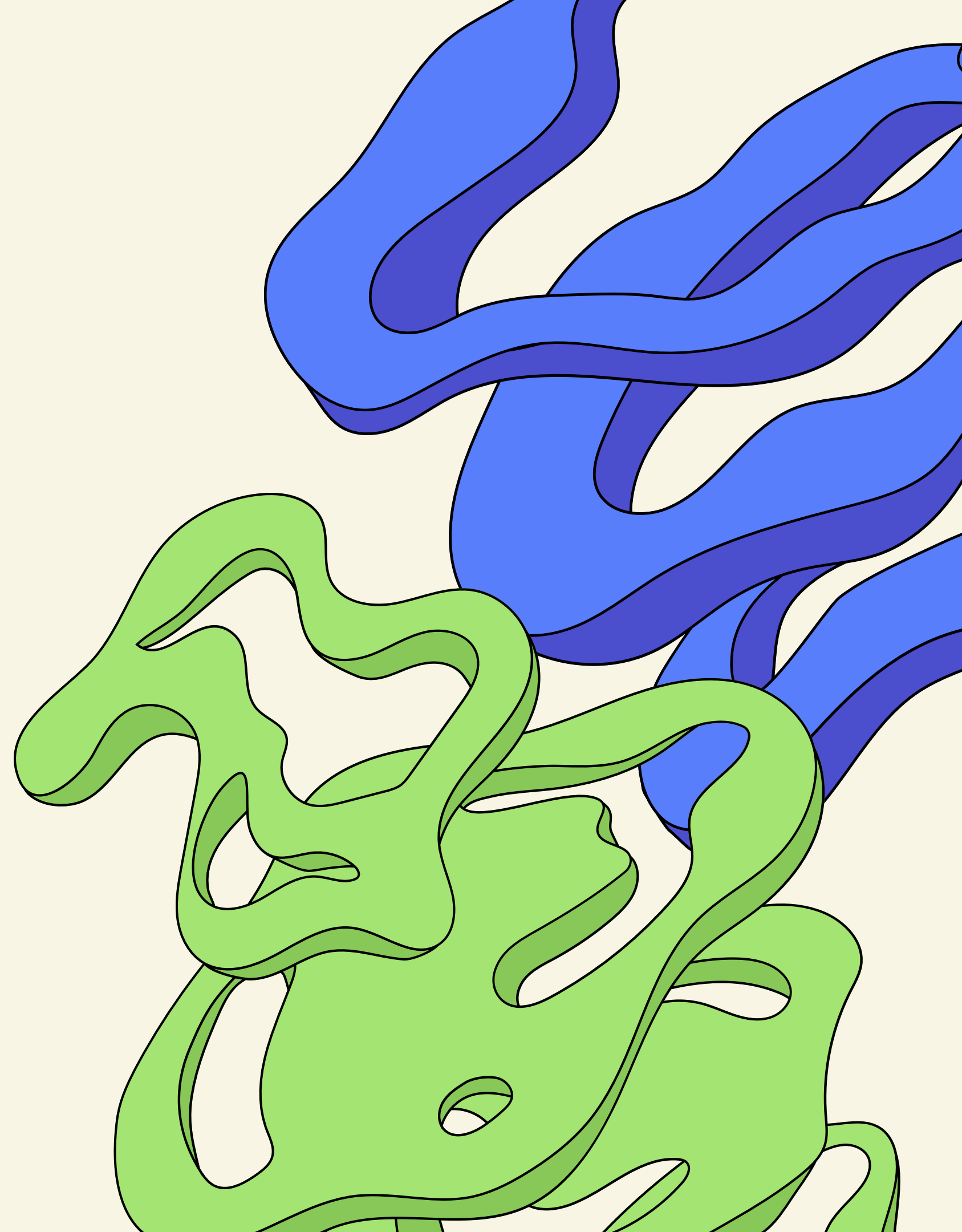
Davide Tonetti - s334297

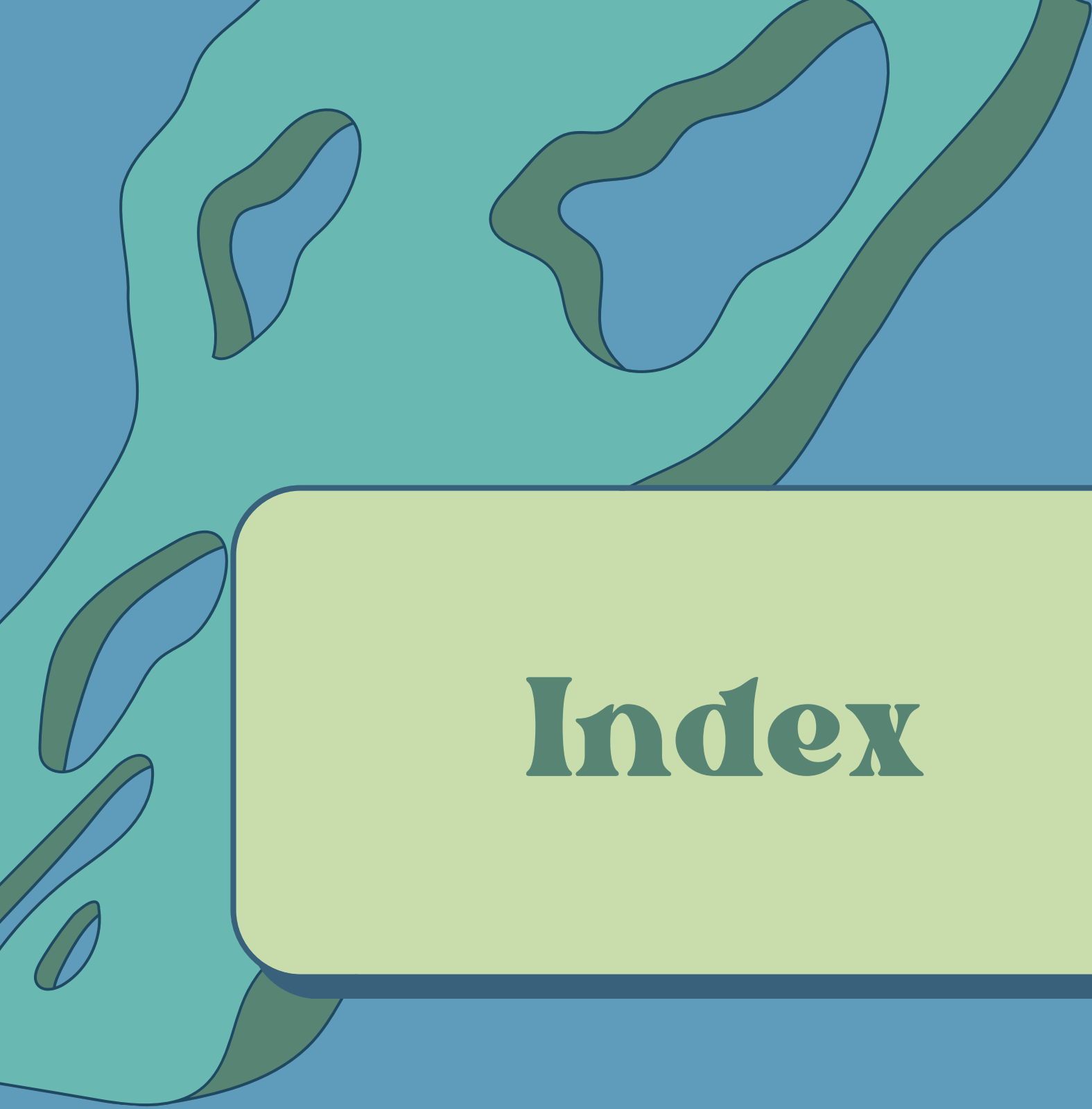
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links
PASSION FOR INNOVATION



A stylized map of the Pacific Northwest region, showing the coastline of Washington, Oregon, and California. The map is rendered in shades of blue and green, with dark green outlines for the coastlines and islands. The word "Index" is centered within a light green rounded rectangle that overlaps the map.

Index

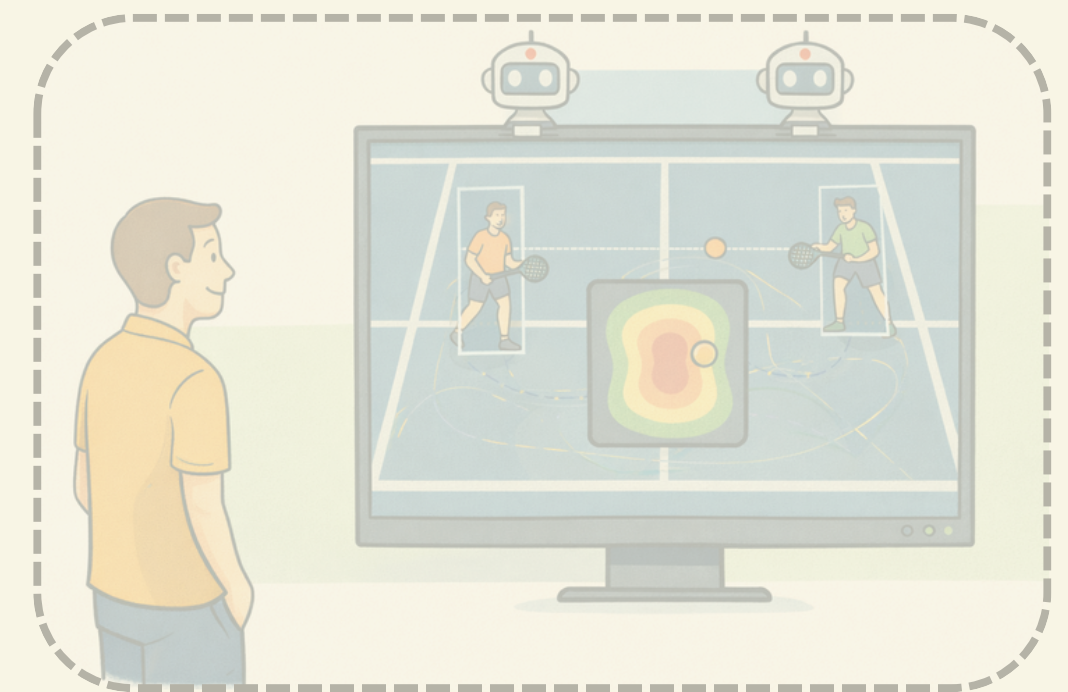
- **Project Evolution**
- **Models**
- **Results**
- **Output Statistics**
- **Possible Developments**

Project Evolution

SET 1: Understanding context, needs, and challenges

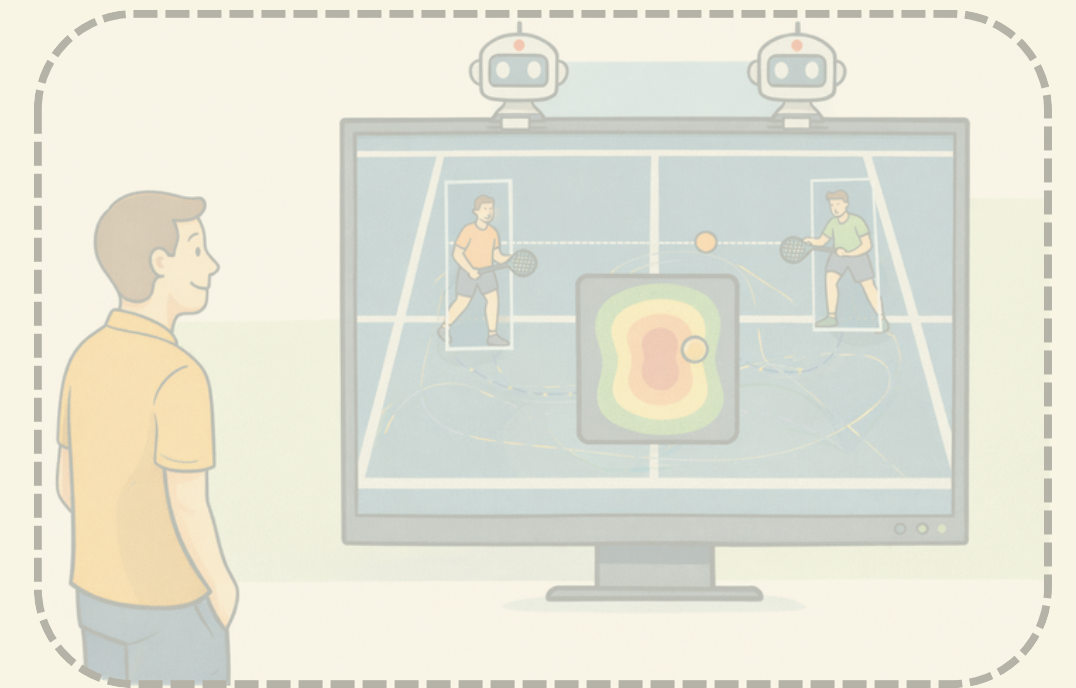


- Defined **project scope** and **target stakeholders**
- Identified **key problems**
- Proposed a **data-driven solution**
- Listed initial **challenges and hypotheses** for testing



Project Evolution

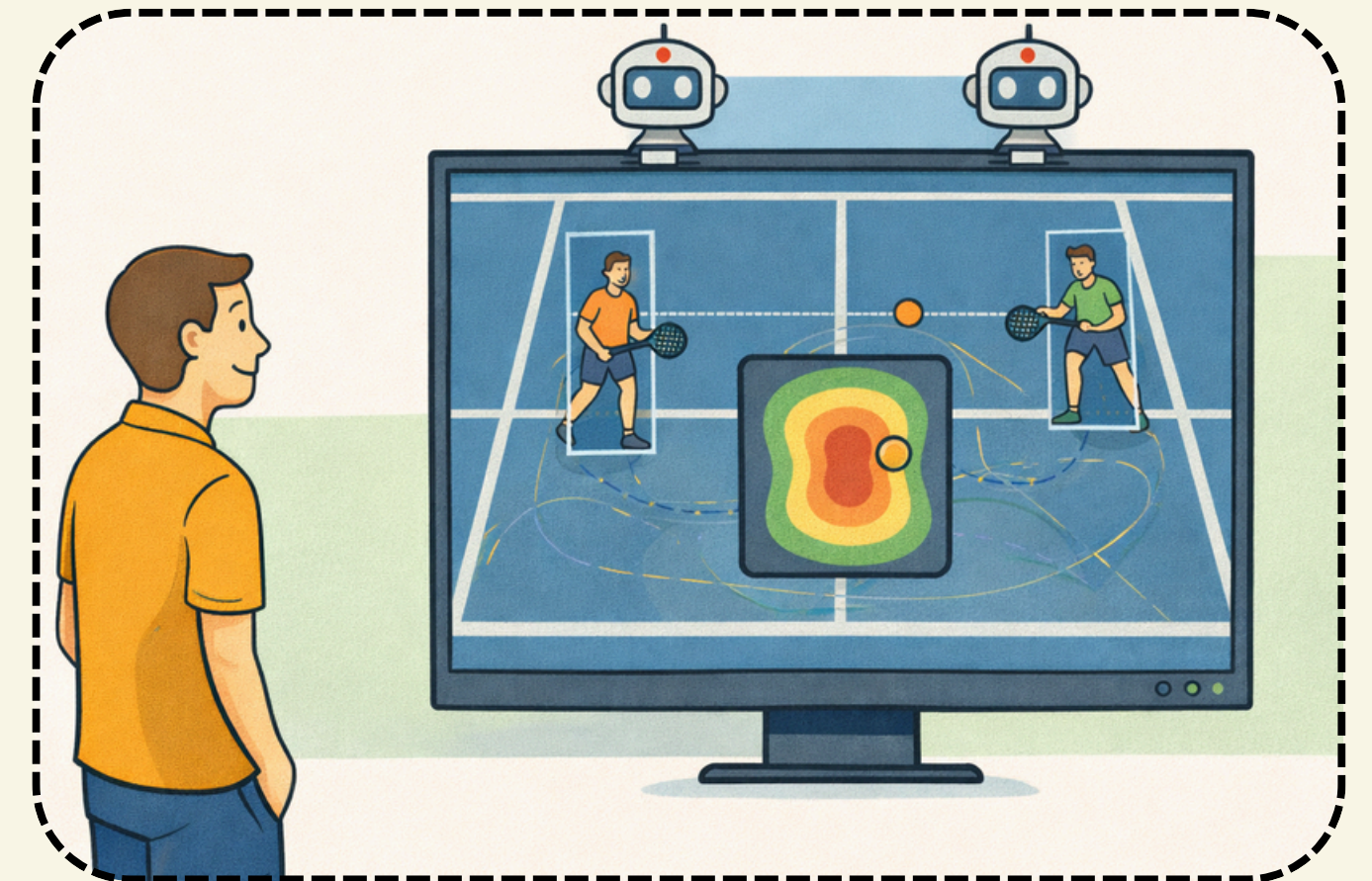
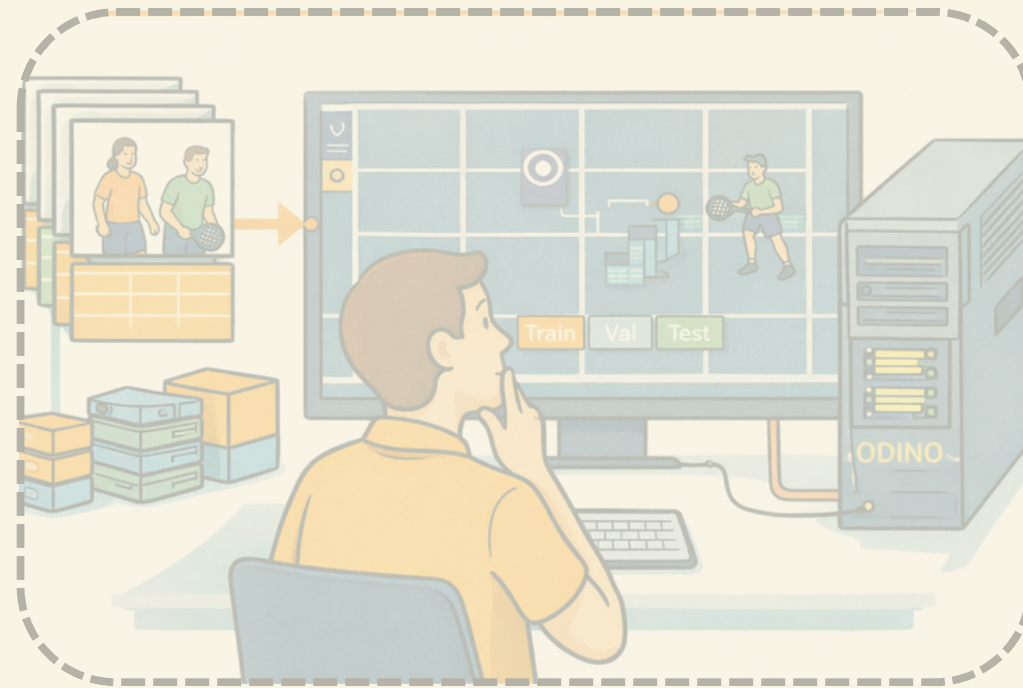
SET 2: Structuring data, tools, and development choices



- Designed **dataset split** at frame level
- Explained choice of **infrastructure**
- Prepared **pipeline for training, inference, and initial analytics**

Project Evolution

SET 3: From models to measurable performance



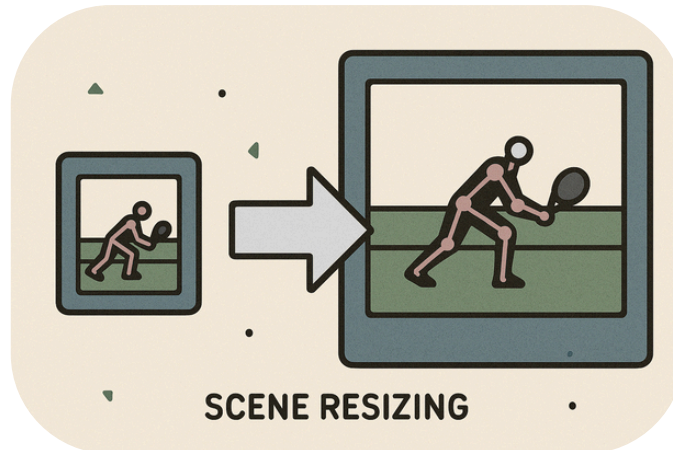
- **Trained** and **evaluated** detection models
- Assessed **precision** and **recall** performance
- Extracted **actionable** outputs
- Observed **model** limitations
- Established **experimental baseline** for further **refinement**

Models



yolo 11n
yolo 10n
yolo 9t

model comparison



640
960
1280

imgsz hyperparameter

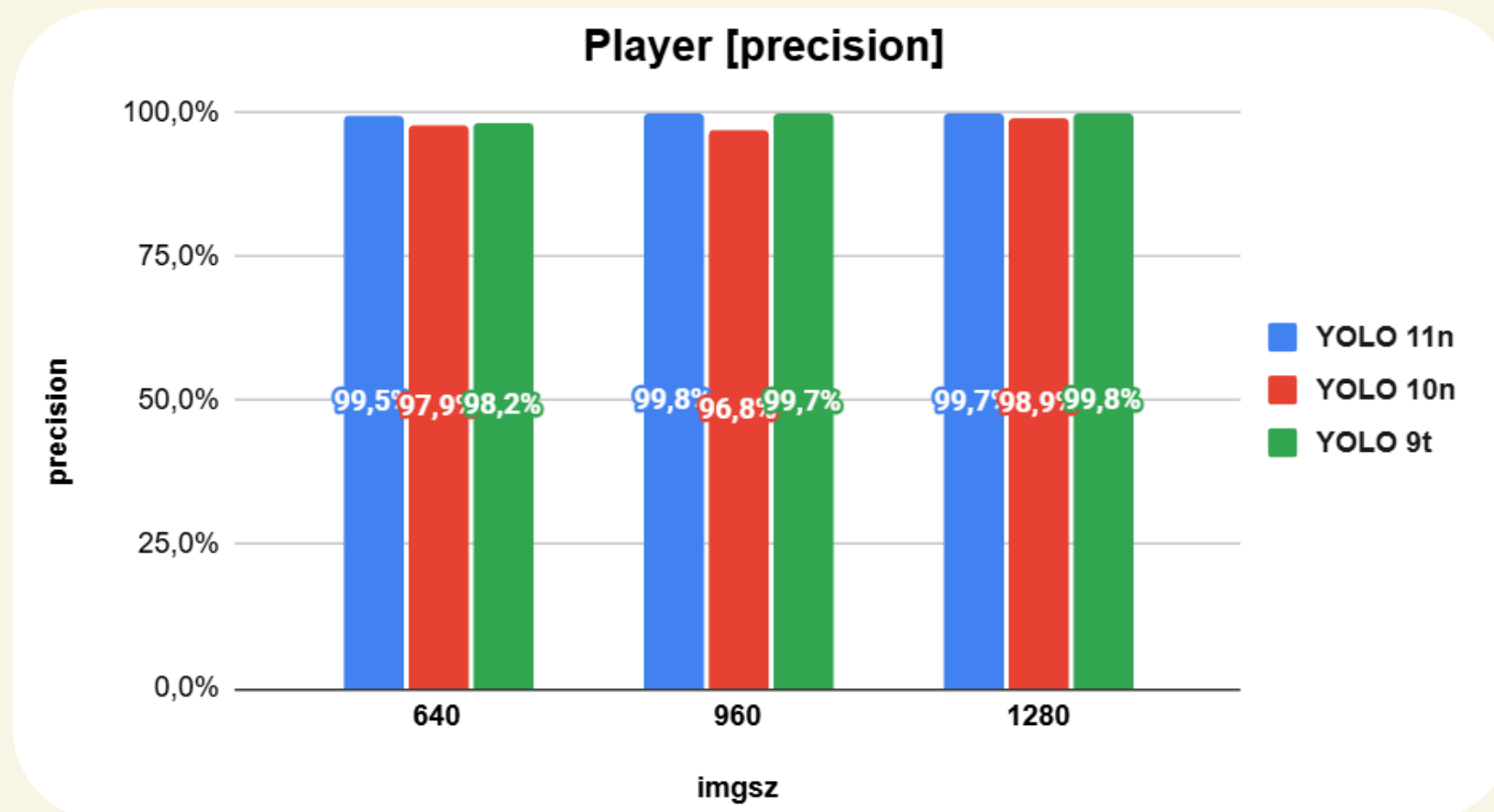
Recall: how many true positives are detected among all actual positives

Precision: how many of the model's positive predictions are correct

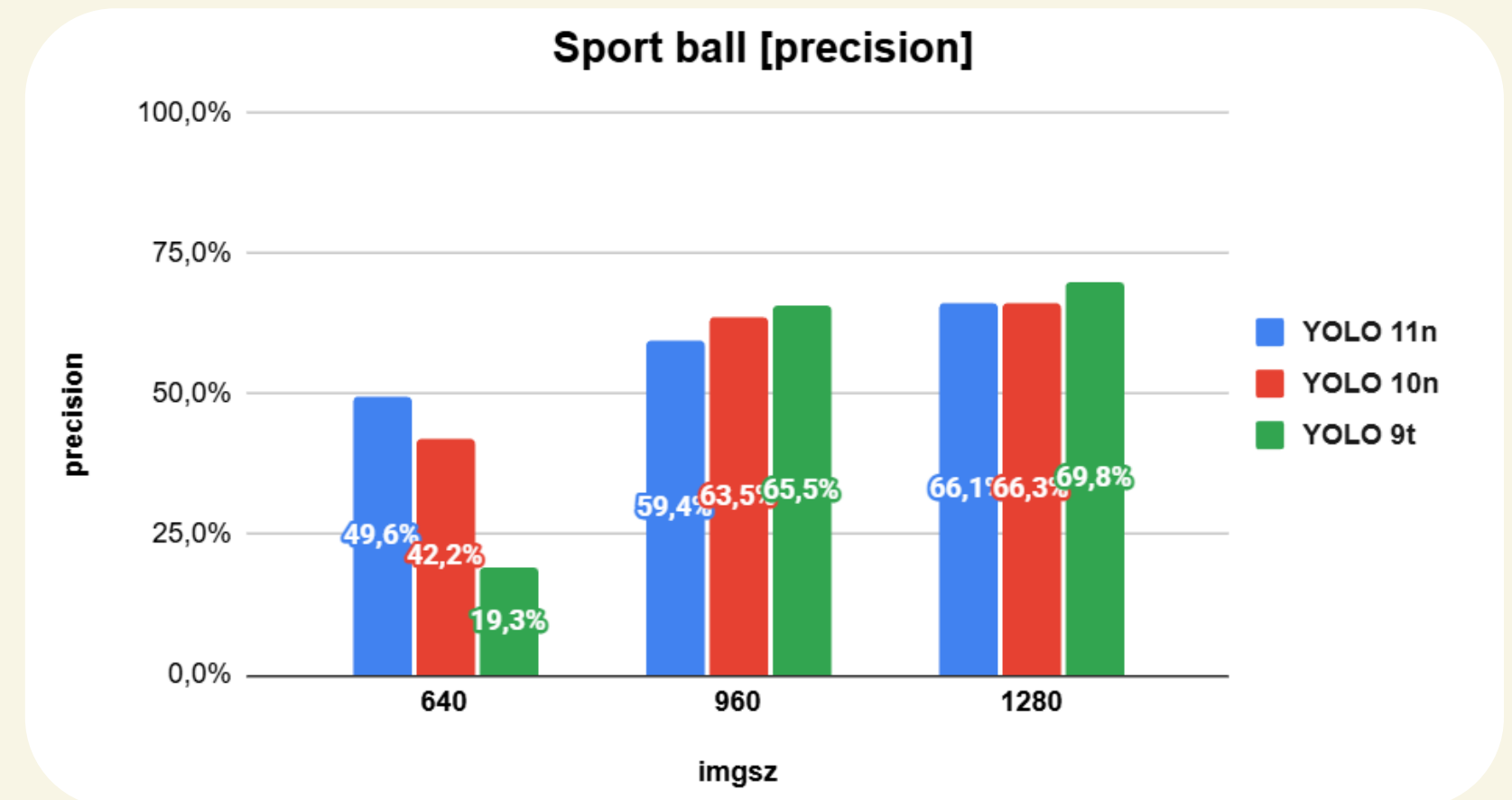
Inference time: Time required to run the inference process on the test set

metrics

Larger imgsz, better precision?

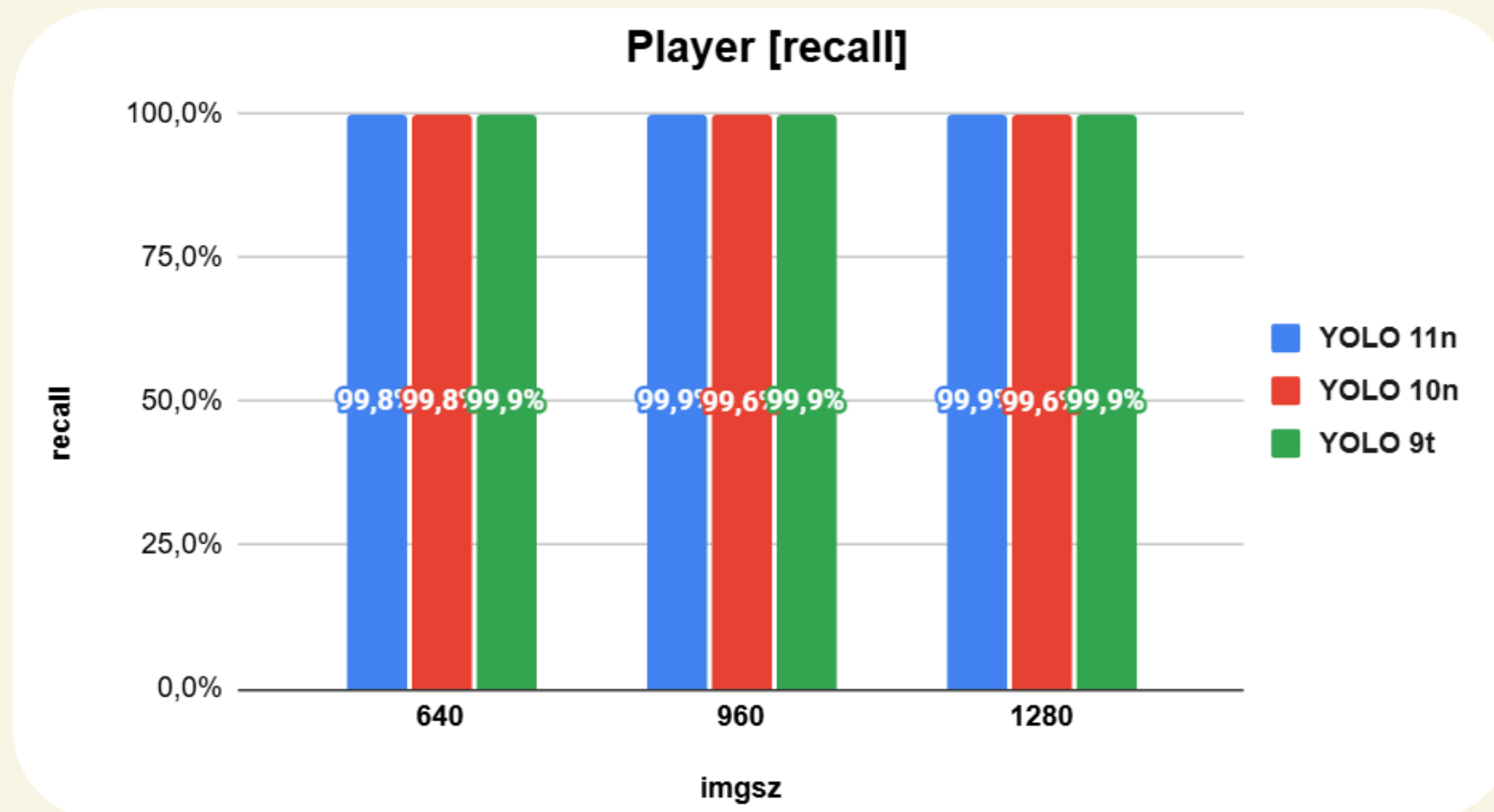


- Similar performance for all configurations
- High precision for all configurations (>97%)

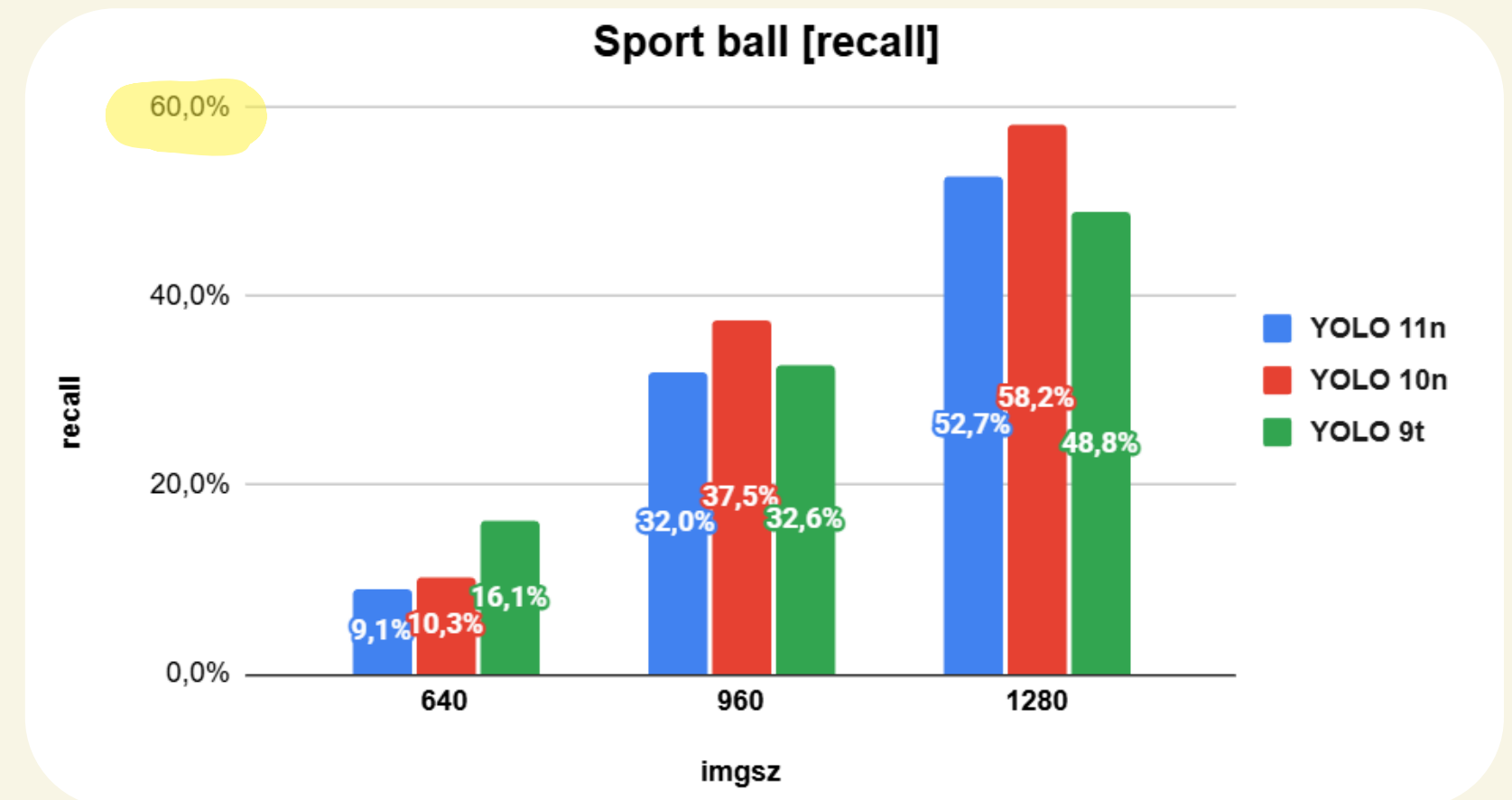


- Precision increases as the image size grows
- With imgsz = 640, the default one, YOLO11n performs best
- Best configuration is Yolo9t with imgsz = 1280

Larger imgsz, better recall?



- Similar performance for all configurations
- High recall for all configurations (>98%)



- Best configuration is Yolo10n with imgsz = 1280
- Recall increases as the image size grows

More training epochs, better results?

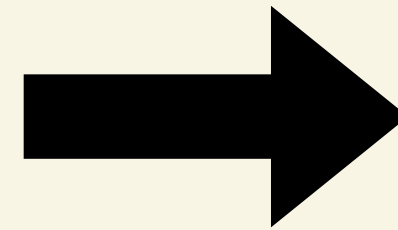
Selected Models

- **Model A | Highest precision & recall**

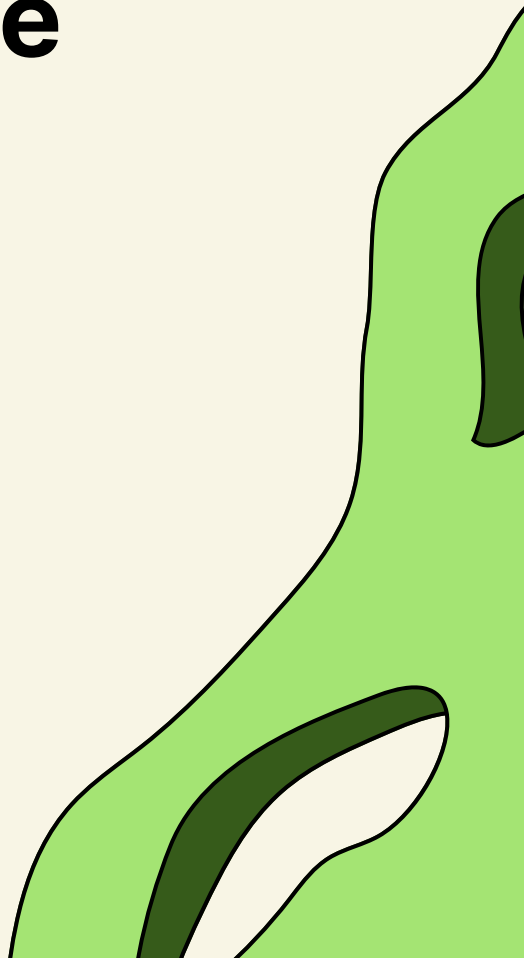
Considering only performance, the best configuration is **yolov10n with imgsz=1280**

- **Model B | Best speed-accuracy trade-off**

Considering both performance and inference time, the best configuration is with **yolov10n and imgsz=960**



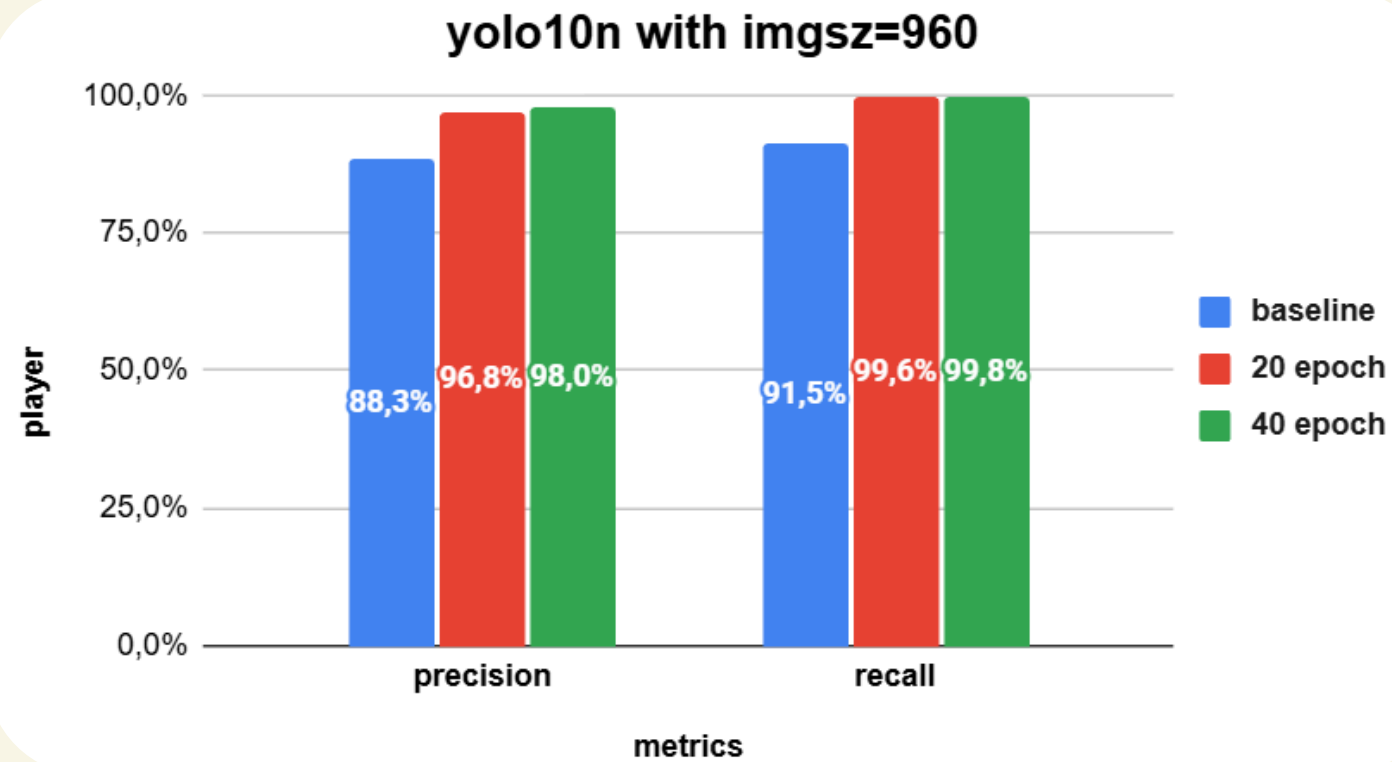
Now train for 20 more epochs and compare the results



More training epochs, better results?

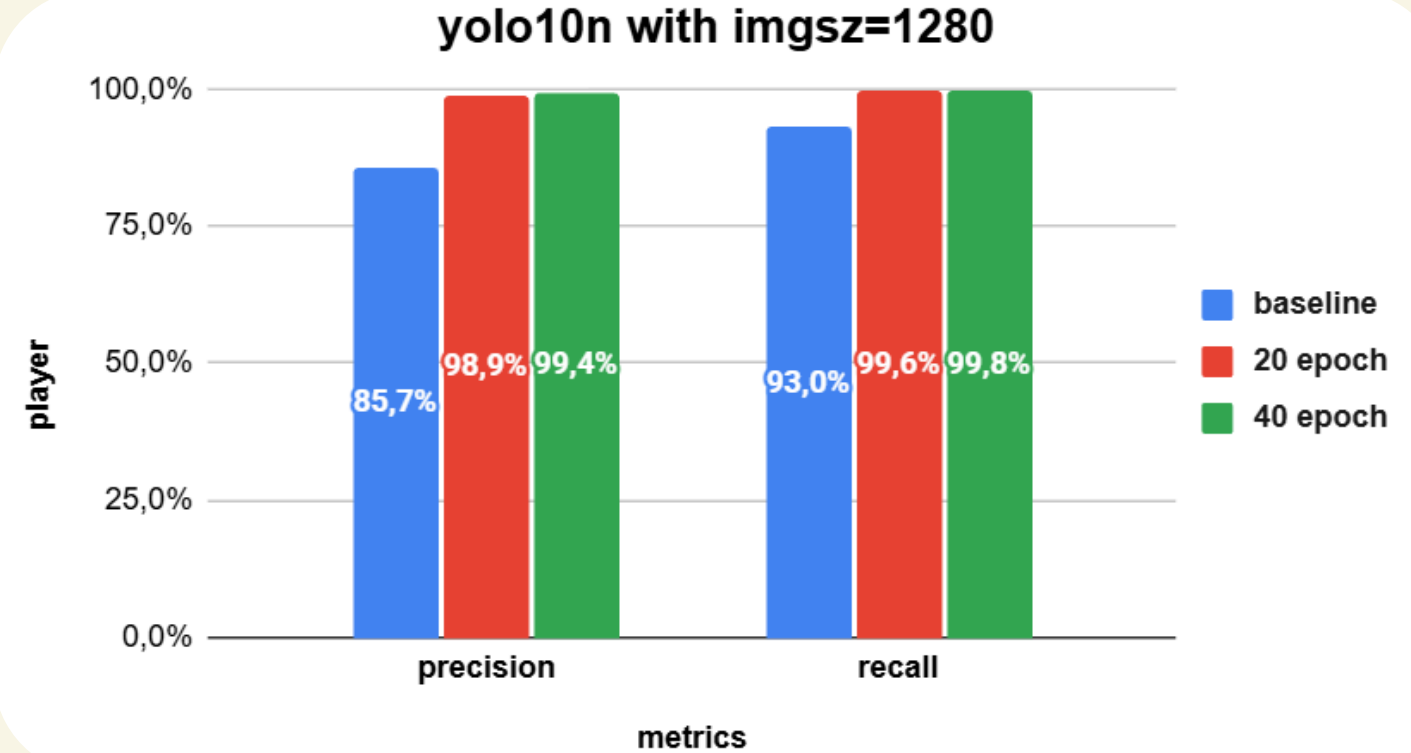
yolo10n with imgsz=960

player

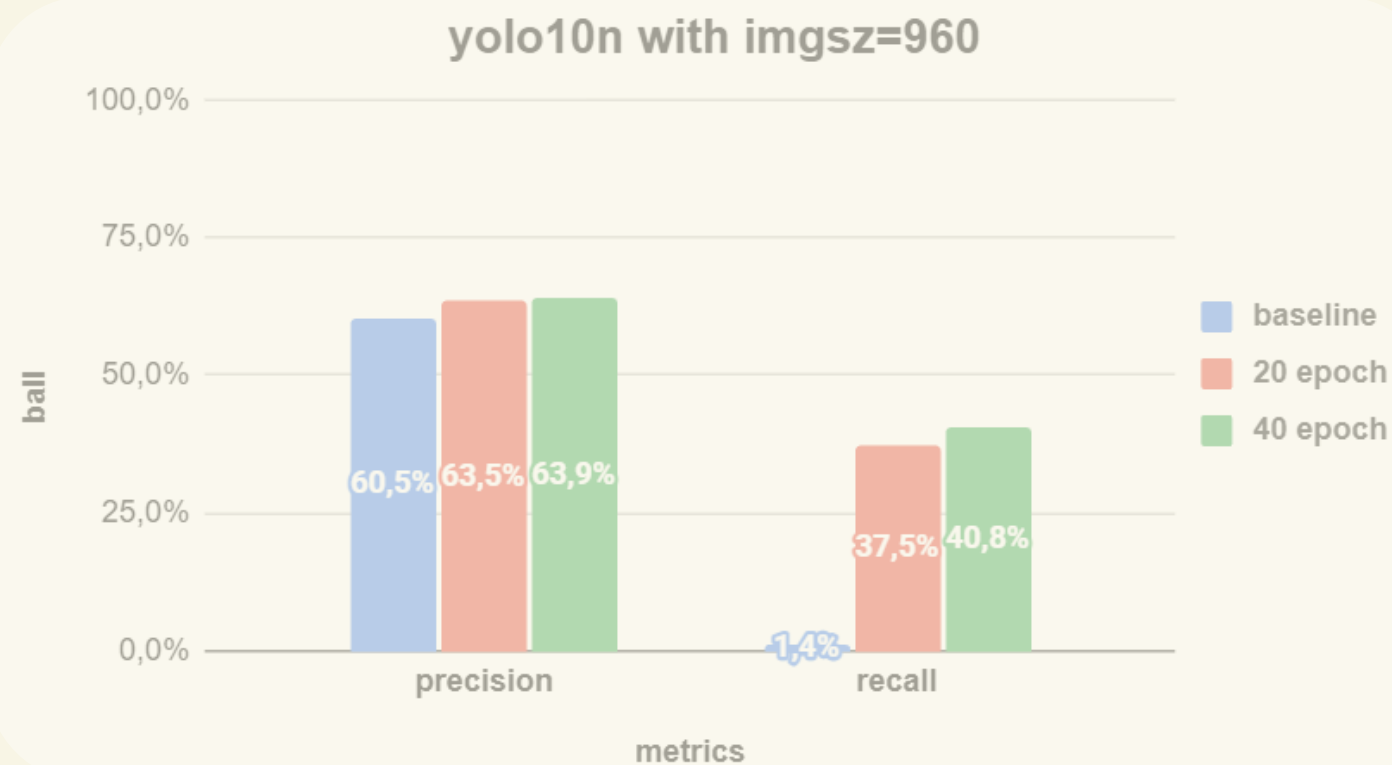


yolo10n with imgsz=1280

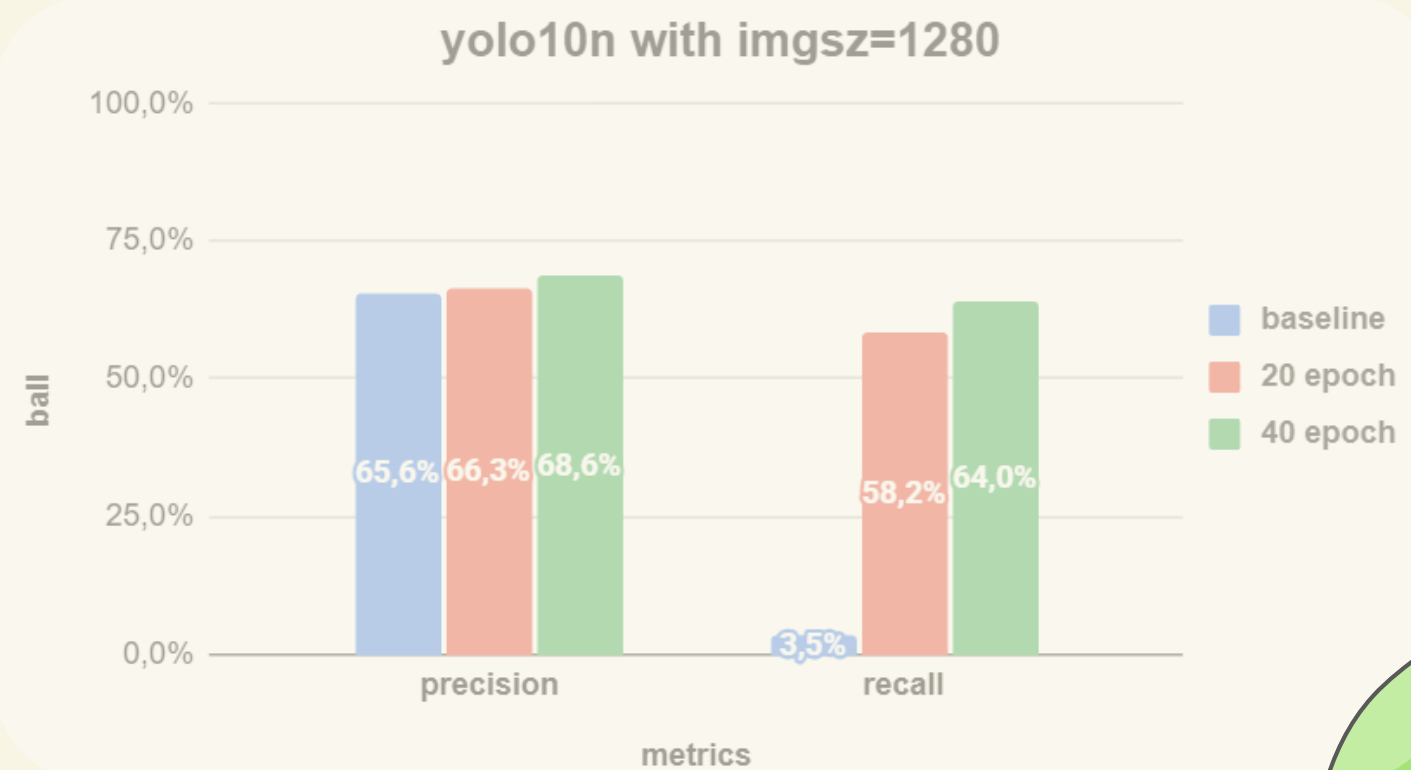
player



ball



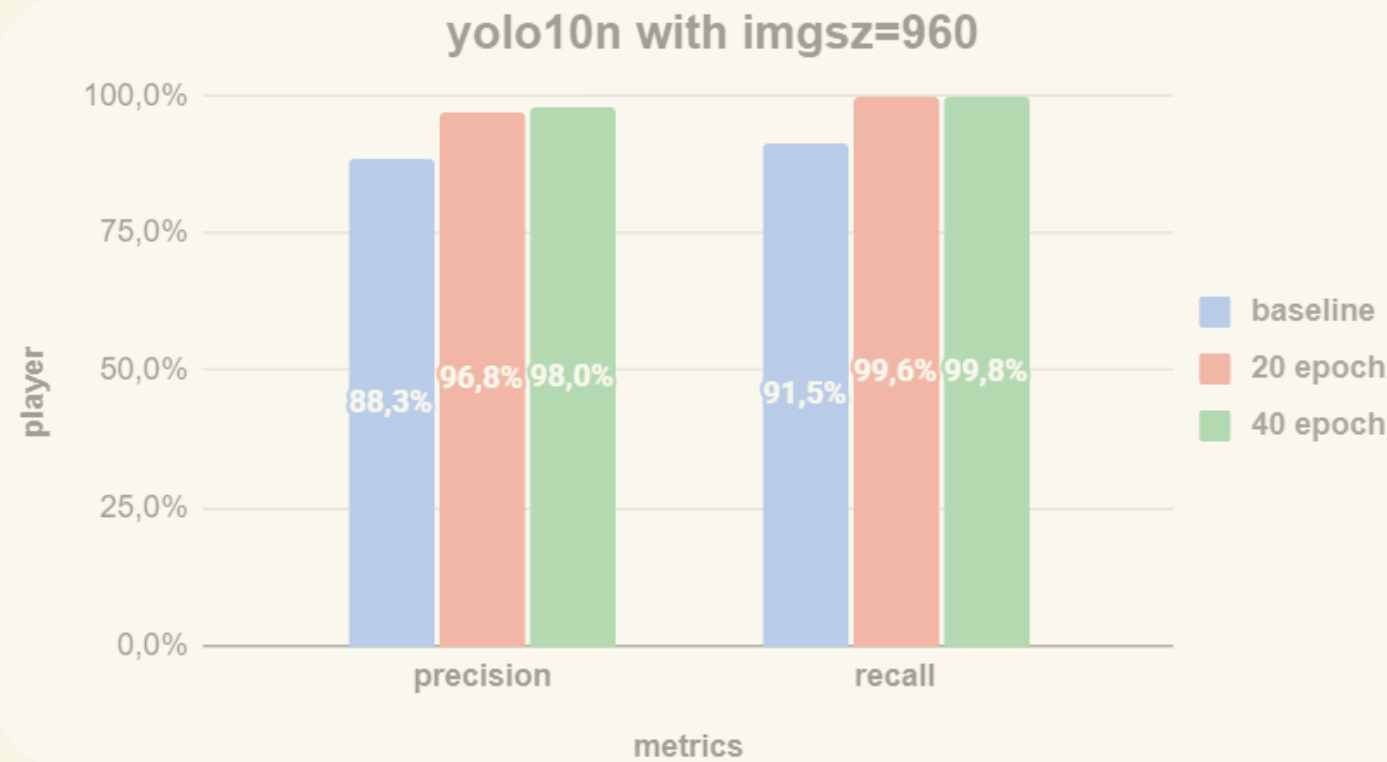
ball



More training epochs, better results?

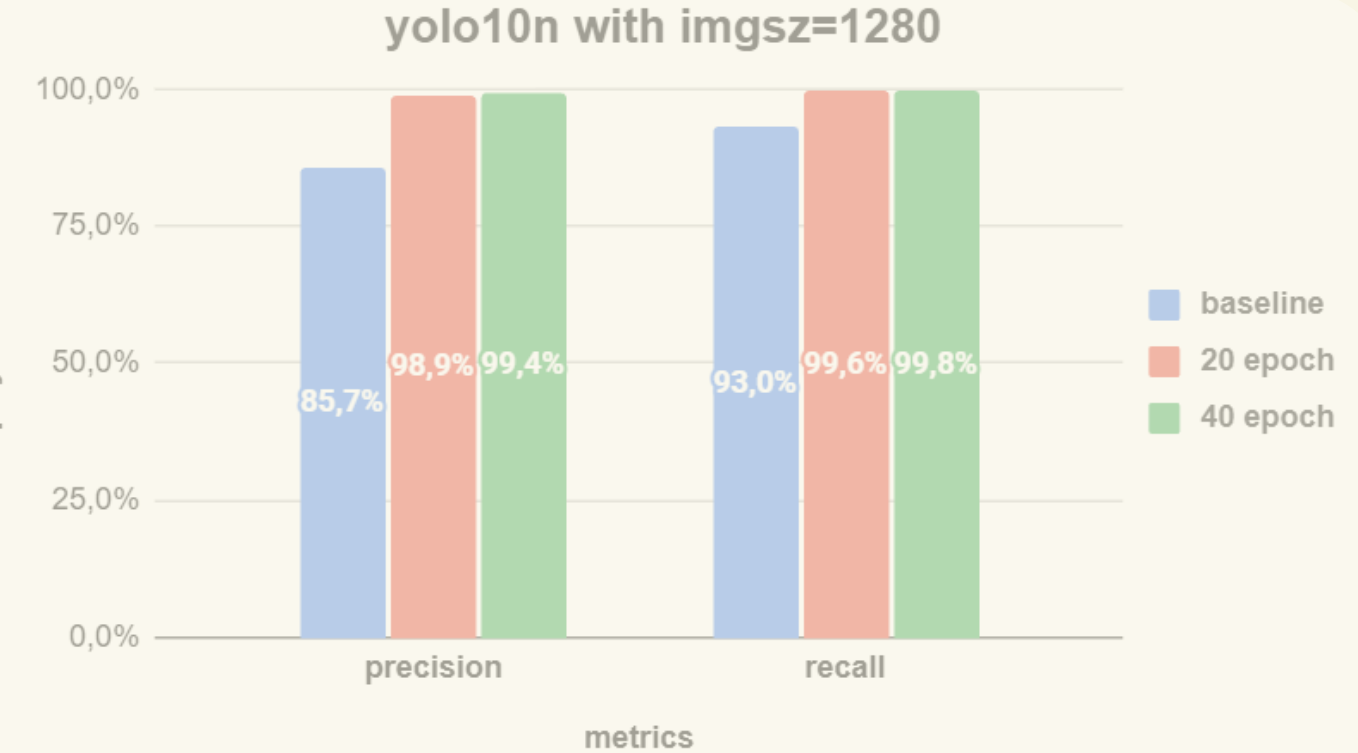
yolo10n with imgsz=960

player

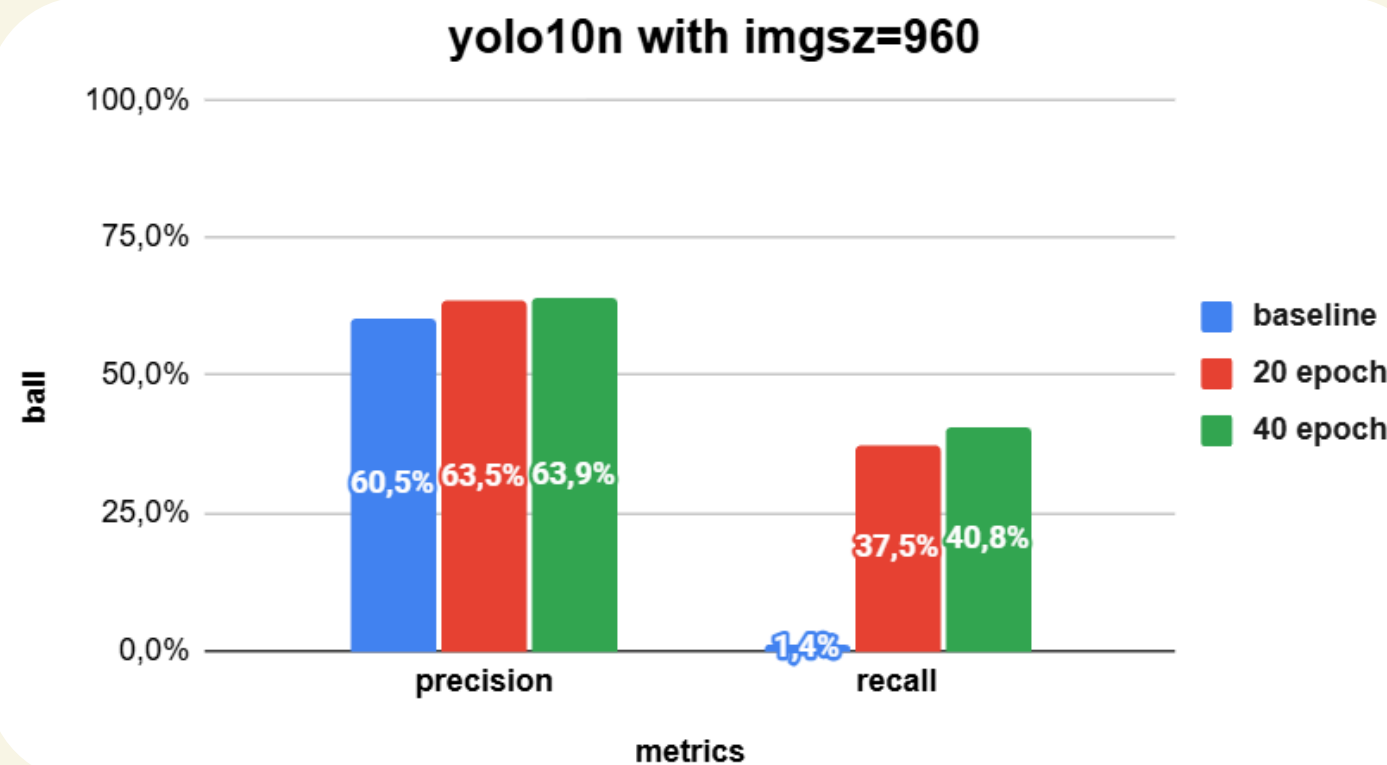


yolo10n with imgsz=1280

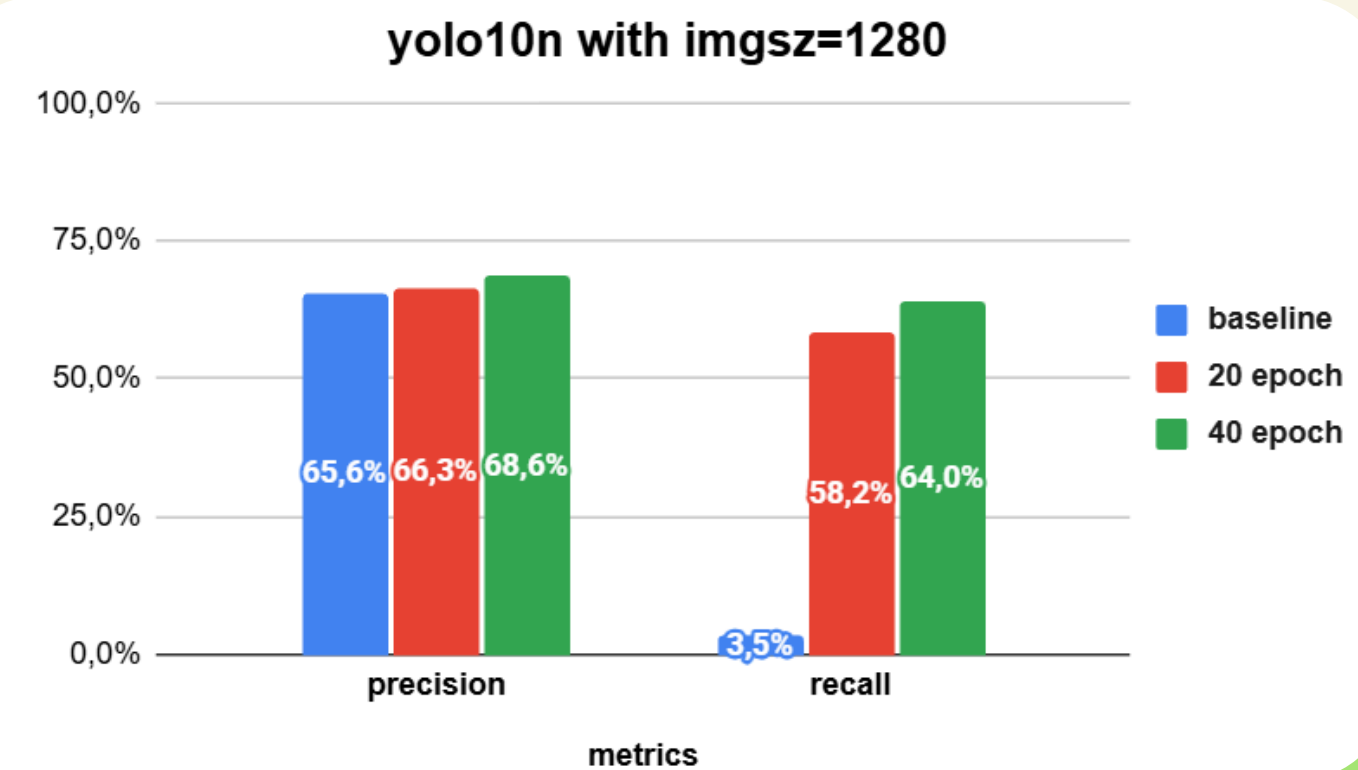
player



ball



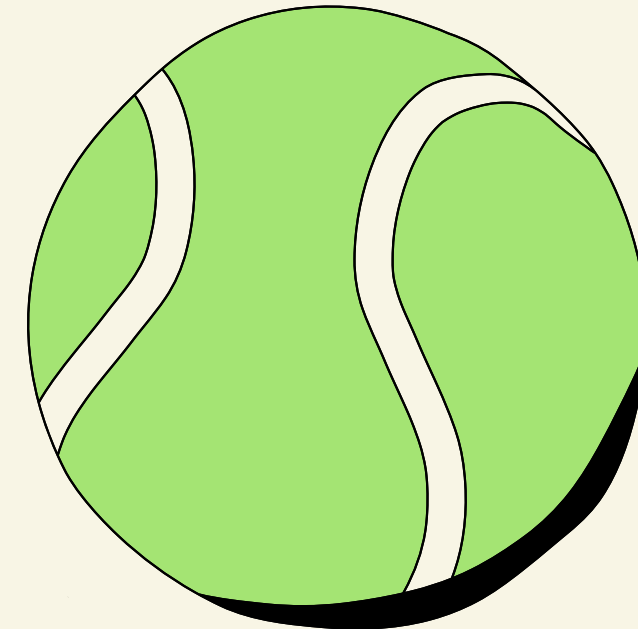
ball



Performances

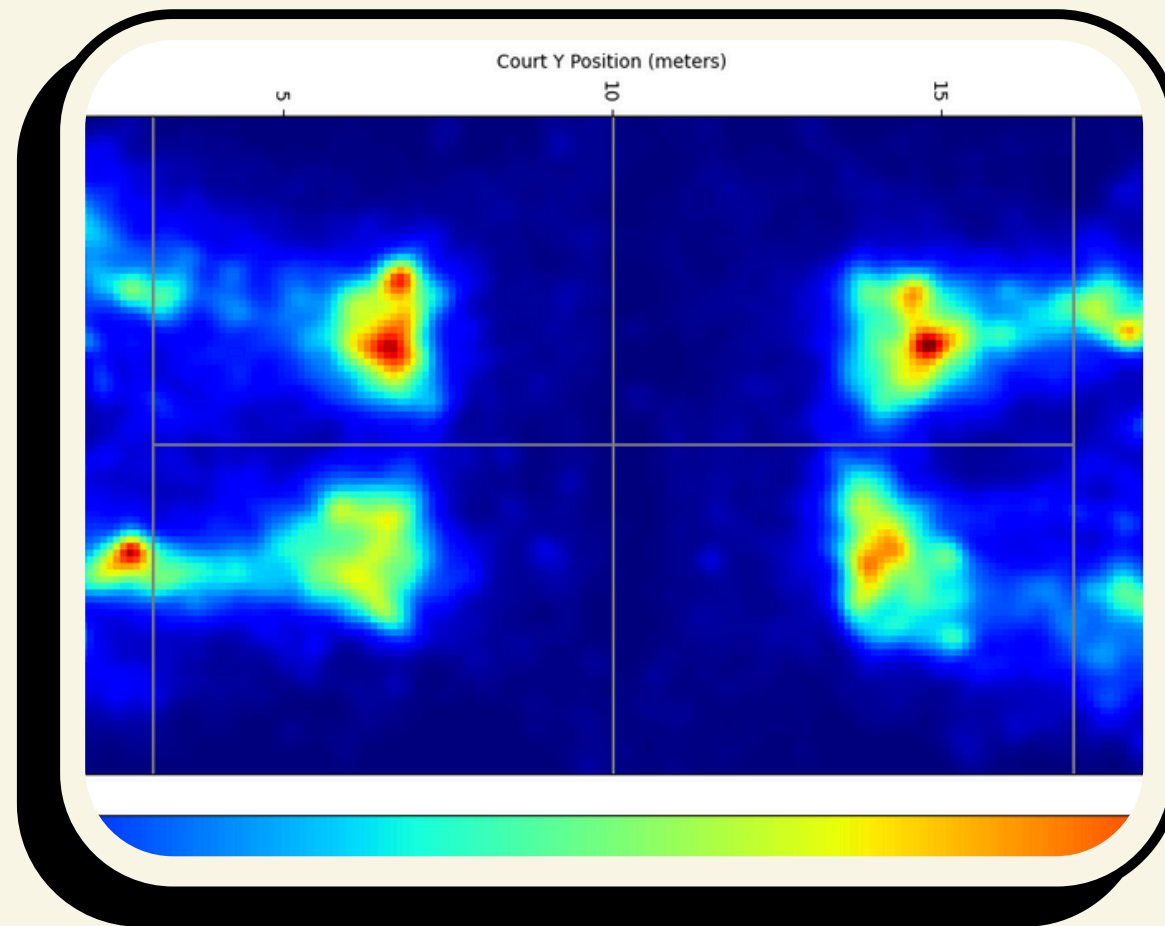


We achieved **near-perfect results** in precision and recall, demonstrating that the model successfully learned to distinguish players from the background.

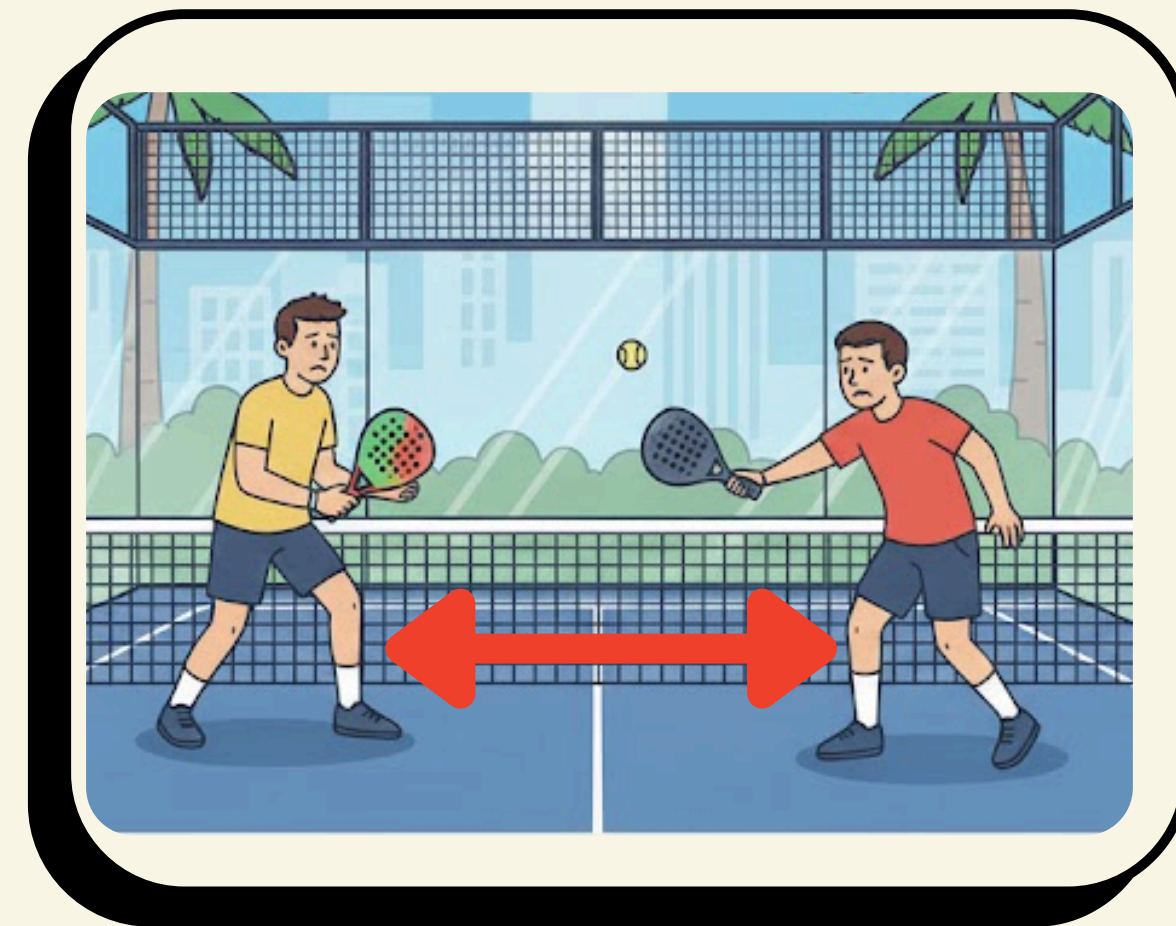


Precision remained relatively stable, while **recall** showed significant improvement. The model now detects the majority of balls in the padel match.

Output Statistics

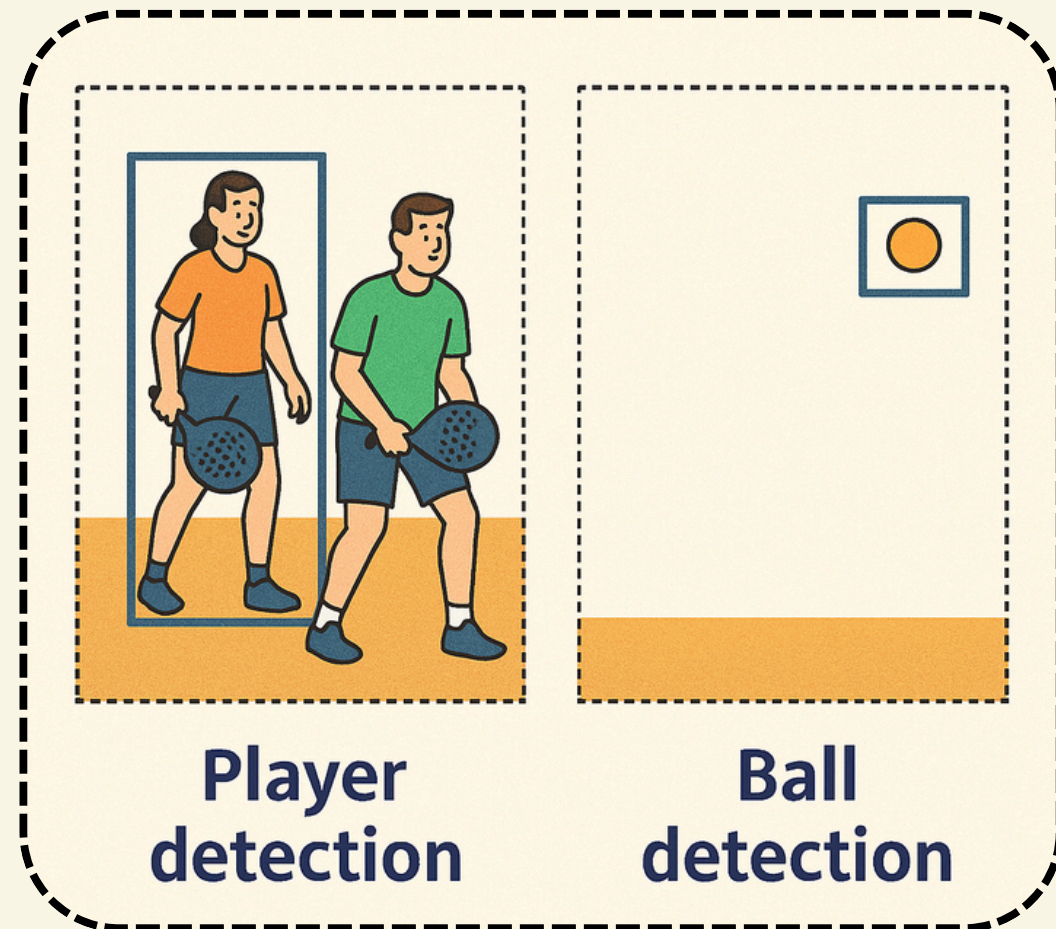


Heatmap



Avg. Distance

Development of Output Statistics



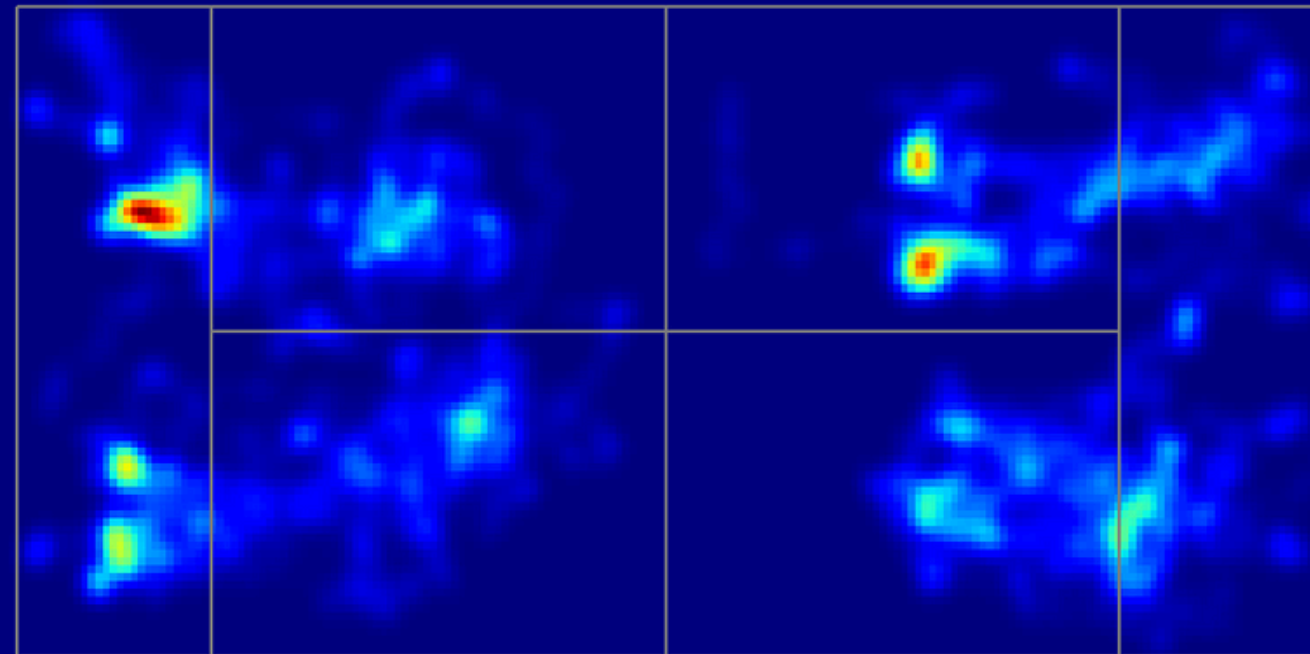
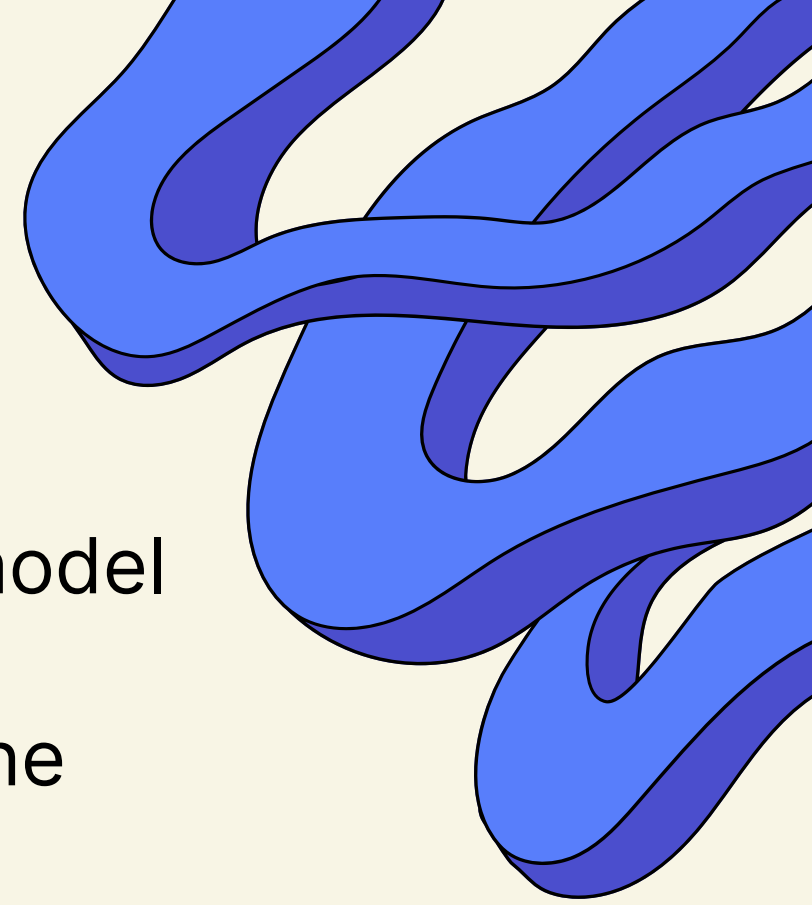
The main aspect of the process of statistic development we can act on is the Bboxes selection from the model's output.

The question we asked ourselves is:

Does using different techniques of bboxes selection actually make a difference in the quality of output statistics?

Raw Bboxes Extraction

The first technique we used is to simply take all the bboxes the model outputs as correct and produce the output directly.
This is a very straightforward method that could be prone to the propagation of errors made by the detection model.



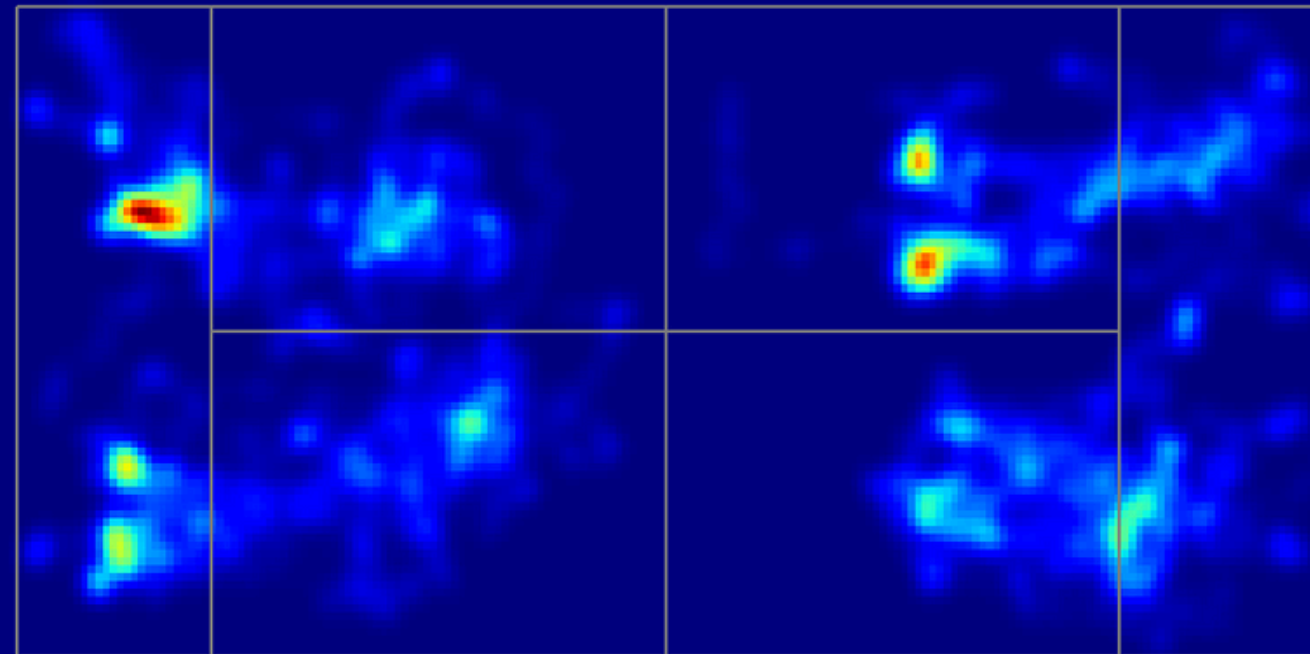
This is the heatmap obtained using the fine-tuned version of the Yolo 10n model over the first set of the female match in the dataset.

Also the model yields an average horizontal distance between players of the same team of: 4,385m

Filtered Bboxes Extraction

The second technique we used is to take all the bboxes the model outputs, project them onto the field and select only the ones which actually fell into the pitch, using them to produce the outputs.

This is still a simple method that solves the problem of external figures classified as players but still doesn't address the missing detections problem.



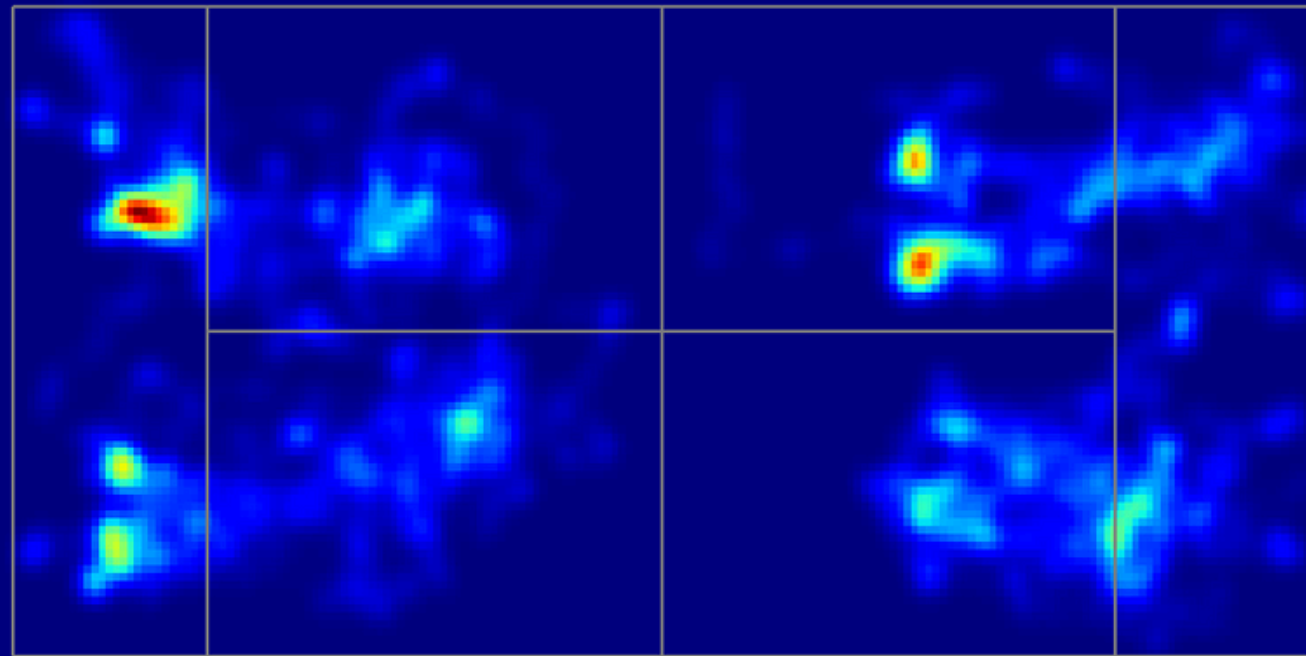
This is the heatmap obtained using the fine-tuned version of the Yolo 10n model over the first set of the female match in the dataset.

Also the model yields an average horizontal distance between players of the same team of: 4,385m

Bboxes Extraction with Tracking

The third technique we used is to implement a tracking system that stores previous positions and momentum of the players and therefore is able to produce an estimate of the player's position whenever the model fails to locate it.

This method implements the filtering technique seen before and also recognize when a rally ends and resets the memory of previous positions and player's momentum.



This is the heatmap obtained using the fine-tuned version of the Yolo 10n model over the first set of the female match in the dataset.

Also the model yields an average horizontal distance between players of the same team of: 4,384m

Conclusions

Although we **lack objective metrics** to evaluate the goodness of our output we have made some final considerations.

The following data represents the number of bboxes discarded/estimated using the filtering/tracking methods over the total number of bboxes provided by the model:

Filtered method:

- 0.0021% over the whole female match
- 0.0% over the first game

Tracking method:

- 0.0604% over the whole female match
- 0.0585% over the first game

B.E.V. Output



In accordance with our project tutor we decided to produce the Bird's Eye View video as an output of our model. This in the video on the right is the BEV video of the first rally of the female match.

This statistic also highlights **problems and challenges** with **ball location and projection** onto the pitch.

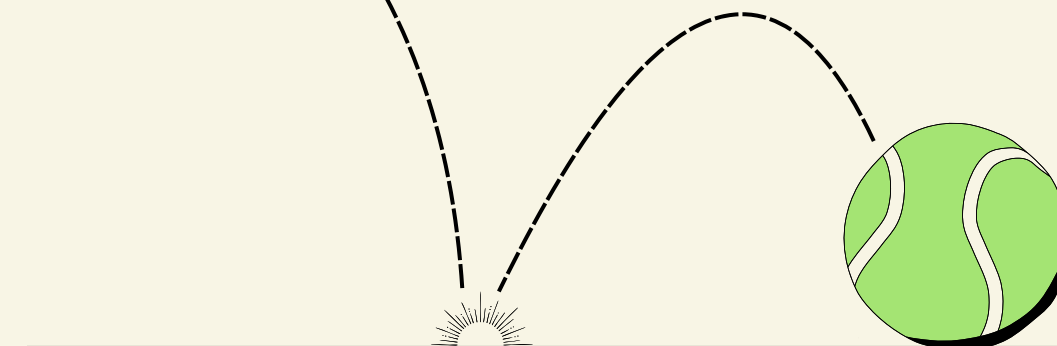
Possible Developments

Ball tracking and Projection:

with an improved dataset and more than one camera angle it would be possible to track the ball and project in onto the field

Shots Statistic: having tracked and projected the ball it would be possible to provide statistics on shots and points

Player Tracking: provided a continuous video it would be possible to track players and handle field switches



Management



Models

Worked on the model's development.



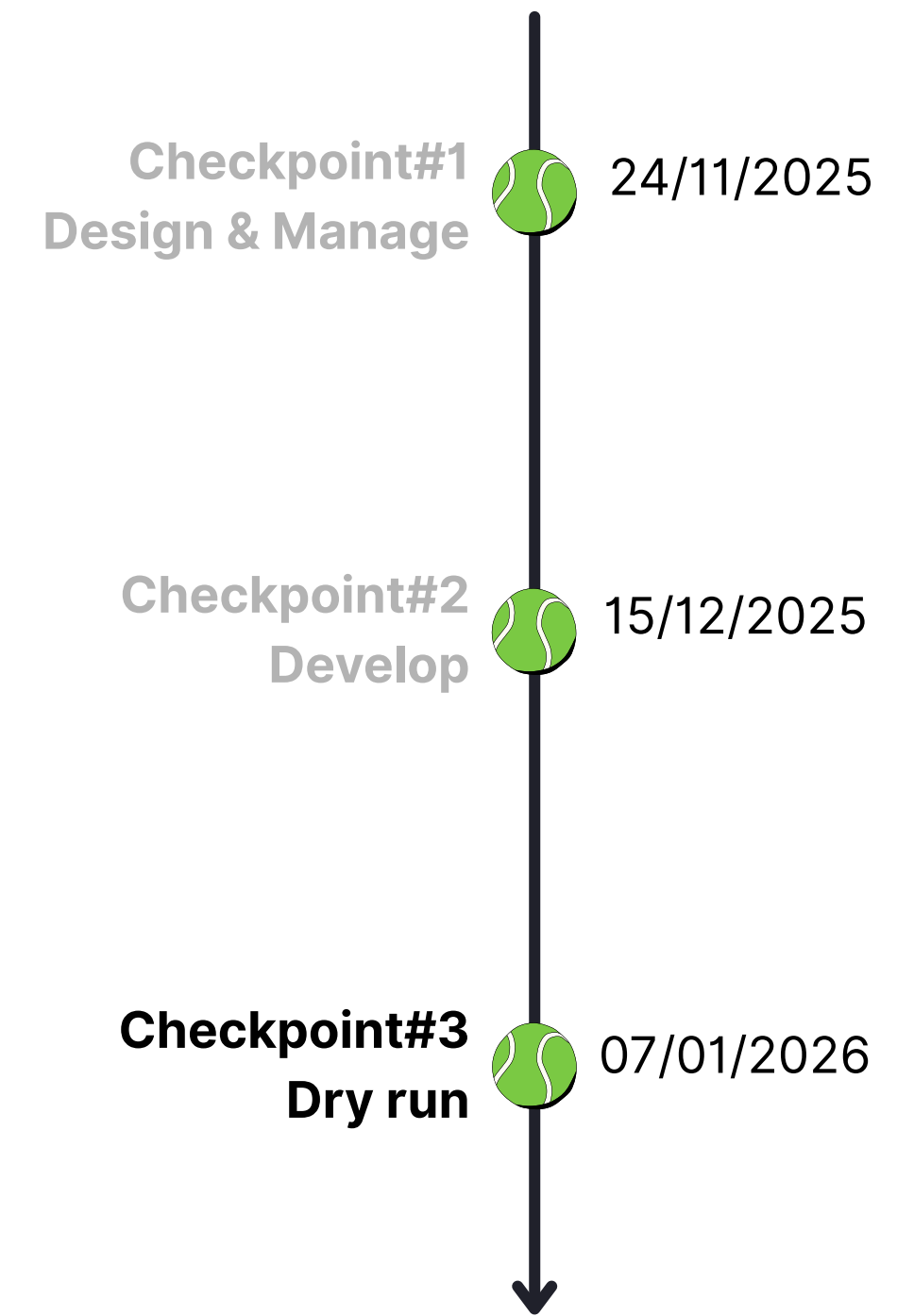
Communication

Took care of the presentation's structure and of the dataset split.



Output statistics

Worked on the statistics and output creation.



Thank you!



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